Software Requirements Specification (SRS)
Project Lane Management System

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1 Introduction

This software requirements specification document will cover all the requirements needed for the Lane Management System project for customer Ayush Agrawal in association with General Motors.

The contents that will be covered in this software requirements specification document are as follows:

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1.1 Purpose

The purpose of the Software Requirements Specification document is to address validation criteria for software along with creating a clear and concise agreement between the customer and developer. This document serves as the foundation of why the software was developed and how it addresses client needs. Additionally, this document outlines how the software is intended to assist drivers who utilize the Lane Management System within their vehicle(s).

1.2 Scope

The Lane Management System is an embedded automotive software product that is composed of the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System.

The Lane Management System is a product to help assist drivers. The driver is still responsible for operating the vehicle and has the ability to override the corrections. The benefits to the Lane Management System are as follows:

- **Improving driver attentiveness:** The Lane Departure Warning System alerts drivers when they begin to drift outside of lane boundaries.
- **Preventing lane drifting caused collisions:** The Lane Keeping System will make steering adjustments when the vehicle is exiting the lane and the driver is unattentive, helping to bring the vehicle back into the correct lane.
- **Reducing driver fatigue:** The Lane Centering System takes over the repetitive task of maintaining lane position, making constant small adjustments to keep the vehicle centered.

The Lane Management System operates on vehicles that travel on roads that are properly marked as determined by the system. It determines that they are marked properly by using imagery from the vehicles cameras. The system then determines the vehicles position within the lane and warns the driver if there is a lane departure, and steers the vehicle back into the lane if driver does not respond and make a correction themself. The system also applies minor corrections to ensure the vehicle is near the center of the lane. The purpose of the Lane Management System is to correct unintentional lane departures and keep the car within a certain distance of the center of the lane.

The Lane Management System does not interact with speed, braking, or acceleration and doesn’t change lanes. The Lane Management System also does not detect people, obstacles or other vehicles. The Lane Management System is also only operable between 35 and 85 mph while also detecting properly marked lanes. If those conditions are not met then the system deactivates and notifies the driver.

1.3 Definitions, acronyms, and abbreviations

Camera Sensor Subsystem: The collection of cameras on the car that capture images and send them to the image processing subsystem.

Image Processing Subsystem: Processes received images and identifies lane markings.

Lane Centering System (LCS): Makes small adjustments to the steering wheel in order to keep the vehicle centered in the lane.

Lane Departure Warning System (LDWS): Displays visual and auditory warnings to the driver through the user interface when the vehicle leaves the lane.

Lane Keeping System (LKS): Makes larger adjustments to the steering wheel in order to stay within the lane. Considered an enhancement of LDWS.

Lane Management System (LMS): The software as a whole, with multiple systems and subsystems included within it.

Path Prediction Subsystem: Uses the information gathered by the Image Processing Subsystem and the Vehicle State Estimation System to predict the path of the vehicle and activate the LDWS, LCS, and LKS if necessary.

Supervisory Control System: Controls all of the subsystems, disables and enables them as necessary as well as provides diagnostics.

User Interface System: Displays warnings and notifications as necessary.

Vehicle State Estimation System: Calculates the speed of the vehicle, steering angle, and road curvature.

1.4 Organization

After section 1, the rest of the document is organized as follows:

Section 2 details high level topics relating to the LMS system. Product context, major functionality, user expectations, constraints, and assumptions made about operating conditions are discussed.

Section 3 includes an enumerated list of requirements that define the behavior, constraints, and invariants of the system.

Section 4 models the structure and behavior of the system in accordance with the requirements from section 3. The following models/diagrams are included:

- Use Case Diagram

- Domain Model
- Sequence Diagram
- State Diagram

Section 5 provides a prototype of the system to demonstrate the system behavior.

Section 6 provides resources used to research the domain and design the LMS.

Section 7 provides a point of contact to learn more about this document and project.

2 Overall Description

This section will cover the context for the product, the major functions that the software will perform, the expectations about the user, safety-critical properties and other constraints, and assumptions made about the hardware, software, environment, and user interactions.

2.1 Product Perspective

A key component of being a driver is being able to acknowledge differences in lanes and their purposes on the highway or public roads. However, not every environment that a driver will usually be in makes it easy to identify lane markings on the road. The implementation of a Lane Management System addresses safer driving by detecting lanes and properly adhering to road rules through lane detection. As the automotive industry continues to evolve, new features are being created to ensure safety and security for future drivers which allows for the expansion of Lane Management systems.

The Lane Management System is an umbrella system containing three subsystems of which are integrated to serve as lane management. The subsystems are Lane Keeping System, Lane Centering System and Lane Departure Warning System.

Interface Constraints:

- **System Constraints:** The Lane Management System will have to interact with other systems within the vehicle to be able to fully function. This includes being able to communicate with the vehicle's braking and steering system, along with bridging into other vehicle safety systems the vehicle may have in place outside of LMS.
- **User Interface:** The Lane Management System has to be able to directly notify the driver about changes or warnings within their driving. This may come in different forms but directly ties into the vehicle dashboard for visual warnings and notifications, the vehicle's steering wheel to feel the torque difference when any other subsystem under LMS is activated and the vehicle's audio system to be able to audibly notify the driver about changes or precautionary measures that should be taken.
- **Hardware Interface:** The Lane Management System must interact with the camera sensors within the vehicle to identify lane markings
- **Software Interface:** The Lane Management System works with software to maintain the camera sensor feedback, such as memory usage and data management of the given

feedback

- **Communication Interface:** The Lane Management System ties into the visual and audible notifications given off by the vehicle through the dashboard and audio system, providing communication for the driver

2.2 Product Functions

Lane Detection: The Lane Management System uses the vehicles cameras to capture images and processes them to determine where the vehicle is within the lane if it is deemed that the road markings can be detected properly.

Lane Departure Warning: If the vehicle is leaving the lane then the Lane Departure Warning System warns the driver on the dashboard and by vibrating the steering wheel. The warnings stop once the vehicle returns to its lane.

Lane Keeping: If the driver does not respond to the warning then the Lane Keeping System steers the driver back towards the center of its lane and prevents the driver from drifting into the other lane or off the road as long as the needed torque doesn't exceed the threshold.

Lane Centering: While the vehicle is in its lane, the LCS makes steering adjustments to keep it centered if it is outside the offset threshold.

Driver Override: If the driver uses a turn signal, then the LMS is deactivated and will not warn the driver or make any adjustments to the vehicle's position. Once the turn signal is disabled then the LMS is reactivated.

Logging: Every item the LMS records information or makes a decision the system will log it with a time stamp

2.3 User Characteristics

The user is expected to be qualified and able to operate the vehicle legally and safely. The user is also expected to have read the provided user manual and understand the functionality of the LMS.

2.4 Constraints

The LMS's behavior and operating conditions are constrained by the following:

1. The LMS requires an interface with the following subsystems to operate:
    a. Camera Sensor Subsystem
    b. Image Processing Subsystem
    c. Vehicle State Estimation System
    d. Path Prediction Subsystem
    e. User Interface System
    f. Supervisory Control Systems

2. Subsystems shall have the capability to operate in parallel with one another. For example, the LDWS may be sending lane departure warnings while the LKS is sending steering input to correct the lane departure.
3. All events occurring within the LMS shall be logged, including: vehicle state, speed headings, driving conditions, what triggered the event, what the event was, all warnings, all overrides.
4. The LMS shall only operate at vehicle speed between 35 and 85 mph (exclusive).
5. The LMS shall only operate when a single lane can be detected from a pair of lane markers.

The LMS is a safety critical system. Meaning failure in certain functionalities of the system could incur serious injury or loss of life. To prevent this, additional security considerations have been made for the following properties:

- **Steering**: with the LMS, the vehicle receives steering input from the driver, the LKS, and the LCS. Multiple actors controlling the steering at once could produce dangerous behavior, so a hierarchy over control has been created. The driver has ultimate priority over steering, and any meaningful torque applied to the steering wheel from the driver will override steering input from either the LKS or LCS. Additionally, because the LKS is activated in response to lane departure, it has steering priority over the LCS.
- **Warnings**: when a lane departure has been detected, the driver must be warned sufficiently. To guarantee this, the LDWS warns the driver through three separate methods: visual, aural, and physical (vibration).

2.5 Assumptions and Dependencies

Some assumptions have been made about the hardware in use with the Lane Management System. The first is that the required hardware components will be present in every vehicle where the Lane Management System is implemented. These hardware components include camera sensors, steering wheel, and a turn signal. In addition it is assumed that the needed hardware to display warning signals to the driver is implemented in any vehicle where the Lane Management System is used such as ability to show warnings on the drivers dash board, ability to give haptic warning through the steering wheel and driver seat, and the ability to play auditorial warnings through the car speakers.

Some assumptions have been made about the software of the vehicle in use with the Lane Management System. The first is that the software for other systems such as steering and driver dashboard will have APIs that the Lane Management System will call to send signals to. Additionally, it is assumed that these software systems will work when these API calls are made, if the software systems are down the API calls will notify the Lane Management System as such.

One assumption has been made about the environment in which the Lane Management System is used. It is assumed that for the scope of this project the Lane Management System will operate under American road laws and regulations.

One assumption has been made about the user interactions with the Lane Management System. It is assumed the user is knowledgeable about the workings of the system, knowing how to override the system and turn it off.

2.6 Approportioning of Requirements

While the documentation has specific requirements that are addressed through a working Lane Management System, other requirements will arise that are to be addressed through future versions of the product. This includes understanding how the Lane Management System should react with other vehicles entering the current driver's lane, along with determining the speed threshold for camera visibility to work.

3 Specific Requirements

1. The Lane Management System shall consist of a Lane Departure Warning System, Lane Keeping System, and a Lane Centering System.
2. The Lane Management shall only be activated within 35 and 85 miles per hours (exclusive).
    2.1. Attempts by the driver to activate the Lane Management System when the vehicle is outside of the speed threshold shall display an error message specifying the required speed threshold to the driver.
3. The Lane Management System shall be unable to make accurate decisions under the following circumstances, in these circumstances the Lane Management System shall be disabled.
    3.1. The road is too wide or too narrow.
    3.2. The lane markings are obscured in any way.
    3.3. The lane markings are faded.
    3.4. The lane markings are overly complicated.
    3.5. There are no lane markings.
4. If the Lane Management System is unable to make accurate decisions due to the failure of sensing devices or inaccurate sensing, the Lane Management System shall inform the driver with a visual and auditory alert.
    4.1. The visual warning shall be a notification that appears on the drivers dashboard.
    4.2. The audio warning shall be a beep played by the car's speakers.
    4.3. If the driver attempts to activate the Lane Management System when any of these faults are present the Lane Management System shall display a warning with the relevant diagnostic message on the driver's dashboard.
5. When the Lane Management System is shut down purposely the Lane Management System shall alert the driver with a visual alert.
    5.1. The visual warning shall be a notification that appears on the drivers dashboard.
6. The Lane Management System shall capture images from the cameras every millisecond to capture and/or calculate:
    6.1. The lane markings.
    6.2. The road curvature.
    6.3. The position of the car in the future.

6.4. The car’s location in the lane.

7. The Lane Management System shall keep a log of all decision making data.
    7.1. Logs shall contain the car's position in the lane, car speed, Lane Management System active status, alerts sent and the times they were sent, and commands sent to the steering system and the times they were sent.
8. The Lane Management System shall be able to be overridden by the driver’s steering at any point in operation.
9. The Lane Management System shall be deactivated when either turn signal is switched on.
10. The Lane Management System shall be deactivated when the driver applies 10 N*m or more of torque on the steering wheel.
11. The Lane Management System shall be reactivated when the turn signal is switched off if it was active before the turn signal was switched on.
12. The Lane Management System shall have a way of grading the image capture quality.
    12.1. The Lane Management System shall have a threshold of image quality to pass in order to stay active.
13. The driver shall be able to disable and enable the Lane Management System.
    13.1. The System shall notify and alert the driver with an auditory alert each time the system is enabled
    13.2. The Lane Management System’s current state shall be displayed on the dashboard
14. Lane Departure Warning System
    14.1. When the driver exits their lane they shall receive a warning consisting of a visual, audio, and physical component.
        14.1.1. The visual warning shall be a notification that appears on the drivers dashboard.
        14.1.2. The audio warning shall be a beep played by the car's speakers.
        14.1.3. The physical warning shall be vibration of the seat and the steering wheel.
    14.2. The Lane Departure Warning System shall be able to notify the driver that it is out of the lane after half a second.
    14.3. The Lane Departure Warning System shall turn off the warning once the driver is back in their lane.
    14.4. Three repeated infractions shall result in warning the driver of the danger of driving distracted.
15. Lane Keeping System
    15.1. If no driver action is taken one second after the Lane Departure Warning System activates the Lane Keeping System shall steer the car back to the center of the lane.
    15.2. Lane Keeping System shall prevent drivers from drifting off the road.
    15.3. When the Lane Departure Warning System is inoperable the Lane Keeping System shall be inoperable.
16. Lane Centering System
    16.1. The Lane Centering System shall continuously check lane markings every 0.1 seconds to detect where the car is relative to the lane.
    16.2. The Lane Centering System shall send steering commands to keep the vehicle’s path centered within the detected lane.
17. Cybersecurity
    17.1. The LMS shall authenticate every message received from each other vehicle subsystem and reject any message that fails that authentication.

17.2. The LMS shall reject messages that timestamps don't fit within a one second window of recency.
17.3. The LMS shall validate every steering command against its allowed values before sending it to the Steering Wheel.
17.4. The Lane Management system shall log every rejected message.

4 Modeling Requirements

4.1 Use Case Diagram

Figure 1 shows the use case diagram created for the LMS. The use case diagram was created to show how the separate use cases are divided between the LDWS, LKS, and LCS as well as the actors responsible for each case. In use case diagrams, the actors (stick figures) connect to ovals which represent the use cases. Multiple actors can point to a single use case. These use cases can include, or extend other use cases. Include's are prerequisites that are often repeated between multiple use cases. Extends are exceptions for use cases such as error messages or overrides.

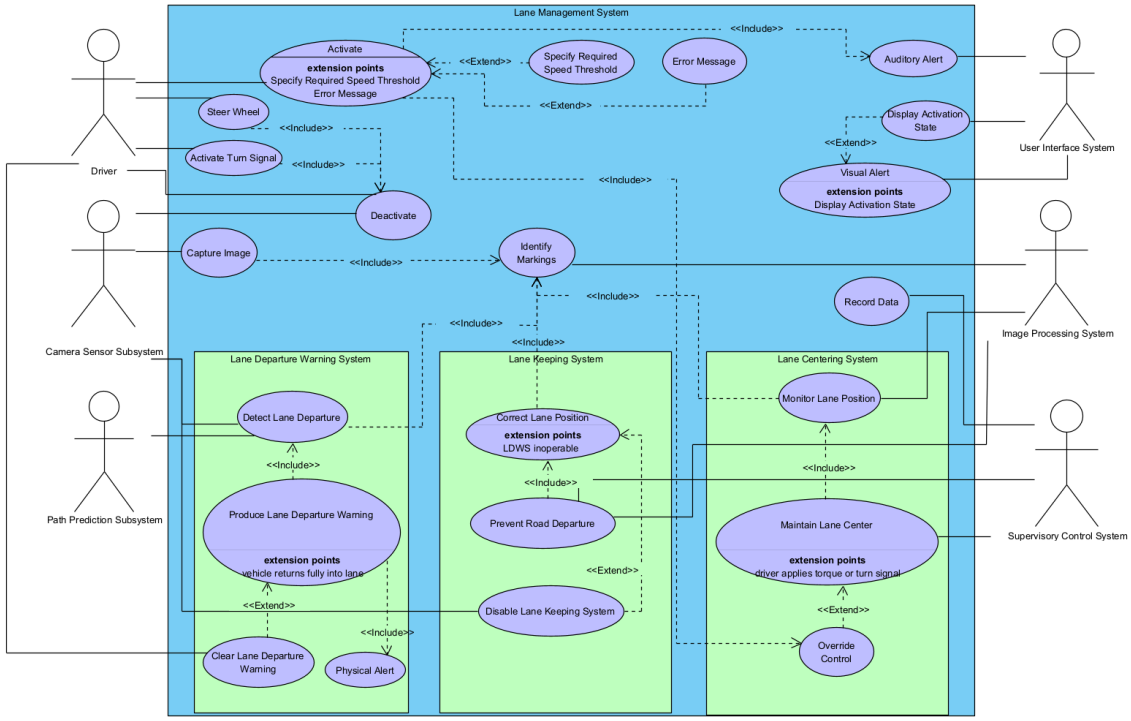


Figure 1: Use case model for the Lane Management System.

Use Cases Descriptions:

Use Case:	Activate
Actors:	Driver
Description:	The driver activates the Lane Management System.

	Perform Error Message if the car's current speed is outside of the required speed threshold. Perform Specify Required Speed Threshold if the car's current speed is outside of the required speed threshold.
Type:	Primary and Essential
Includes:	Auditory Alert
Extends:	N/A
Cross-refs:	2, 2.1, 13
Use cases:	N/A

Use Case:	Deactivate
Actors:	Camera Sensor Subsystem, Driver
Description:	The camera sensors deactivate the Lane Management System if ● The road is too wide or too narrow. ● The lane markings are obscured in any way. ● The lane markings are faded. ● The lane markings are overly complicated. ● There are no lane markings. Additionally, the driver can disable the lane management system. An auditory alert plays following this action.
Type:	Primary and Essential
Includes:	Auditory Alert
Extends:	N/A
Cross-refs:	3, 3.1, 3.2, 3.3, 13, 13.1
Use cases:	N/A

Use Case:	Specify Required Speed Threshold
Actors:	User Interface Subsystem
Description:	Shows the required speed threshold to the driver on the dashboard.
Type:	Secondary and Essential
Includes:	N/A
Extends:	Activate
Cross-refs:	2.1
Use cases:	The driver must have initiated the Activate use case.

Use Case:	Error Message
Actors:	User Interface Subsystem
Description:	Shows an error message to the driver on the dashboard.
Type:	Secondary and Essential

Includes:	N/A
Extends:	Activate
Cross-refs:	2.1
Use cases:	The driver must have initiated the Activate use case.

Use Case:	Identify Lane Markers
Actors:	Image Processing Subsystem
Description:	Identifies lane markers from given images produced by the Camera Sensor Subsystem
Type:	Primary and essential
Includes:	N/A
Extends:	N/A
Cross-refs:	6
Use cases:	Correct Lane Position

Use Case:	Steer Wheel
Actors:	Driver
Description:	The driver adjusts the vehicle's path by turning the steering wheel
Type:	Primary
Includes:	Deactivate
Extends:	N/A
Cross-refs:	10
Use cases:	N/A

Use Case:	Activate Turn Signal
Actors:	Driver
Description:	The driver activates either the left or right turn signal to indicate intent to merge into corresponding lane
Type:	Secondary
Includes:	Deactivate
Extends:	N/A
Cross-refs:	9, 11
Use cases:	N/A

Use Case:	Display Activation State
Actors:	User Interface System
Description:	While the lane management system is active, a symbol is displayed on the screen.
Type:	Secondary

Includes:	N/A
Extends:	N/A
Cross-refs:	13.1
Use cases:	N/A

Use Case:	Produce Lane Departure Warning
Actors:	Path Prediction Subsystem
Description:	A lane departure is detected. An auditory alert, visual, and physical alert is sent to warn the driver.
Type:	Primary
Includes:	Auditory Alert, Visual Alert, Physical Alert
Extends:	N/A
Cross-refs:	14.1
Use cases:	N/A

Use Case:	Auditory Alert
Actors:	User Interface System
Description:	A beep is played over the vehicle's speaker systems
Type:	secondary
Includes:	N/A
Extends:	N/A
Cross-refs:	13.1, 14.1, 14.1.2
Use cases:	Produce Lane Departure Warning

Use Case:	Visual Alert
Actors:	User Interface System
Description:	Display message with relevant information on the screen
Type:	secondary
Includes:	N/A
Extends:	N/A
Cross-refs:	14.1, 14.1.1
Use cases:	Produce Lane Departure Warning

Use Case:	Physical Alert
Actors:	User Interface System
Description:	Produce a physical alert in the form of seat and steering wheel vibration
Type:	secondary

Includes:	N/A
Extends:	N/A
Cross-refs:	14.1, 14.1.3
Use cases:	Produce Lane Departure Warning

Use Case:	Lane Departure Detection
Actors:	Camera Sensor Subsystem
Description:	Determines that the vehicle has crossed lane boundary and has remained outside of the lane within milliseconds, which it then triggers the Lane Departure Warning System to alert the driver
Type:	Primary and Essential
Includes:	Identify Markings
Extends:	N/A
Cross-refs:	14.2
Use cases:	Detect Lane Departure, Physical Alert, Clear Lane Departure Warning

Use Case:	Clear Lane Departure Warning
Actors:	Camera Sensor Subsystem, Image Processing System, Driver
Description:	Turns off the visual, audio, and physical lane departure warnings once the vehicle is fully inside its lane. Additionally, when the turn indicator is on, the lane departure warning system should not alert the driver
Type:	Primary and Essential
Includes:	Identify Markings
Extends:	Detect Lane Departure
Cross-refs:	9, 14.3
Use cases:	N/A

Use Case:	Correct Lane Position
Actors:	Supervisory Control System, Driver
Description:	When the driver takes no action after the Lane Departure Warning System Activates, the Lane Keeping System gently steers the vehicle back to the center of the lane
Type:	Primary and Essential
Includes:	Identify Markings
Extends:	Detect Lane Departure
Cross-refs:	15.1
Use cases:	Identify Markings, Disable Lane Keeping System

Use Case:	Prevent Road Departure
Actors:	Supervisory Control System, Image Processing System
Description:	Applies steering correct to keep the vehicle from drifting out of their lane when the outside of their boundary is the edge of the road
Type:	Primary and Essential
Includes:	Correct Lane Position
Extends:	N/A
Cross-refs:	15.2
Use cases:	Correct Lane Position

Use Case:	Disable Lane Keeping System
Actors:	Camera Sensor Subsystem
Description:	Renders the Lane Keeping System inoperable whenever the Lane Departure Warning System is inoperable
Type:	Secondary
Includes:	N/A
Extends:	Correct Lane Position
Cross-refs:	15.3
Use cases:	N/A

Use Case:	Monitor Lane Position
Actors:	Camera Sensor Subsystem, Image Processing System
Description:	Continuously evaluates the identified lane markings to determine the vehicles lateral position relative to detected line
Type:	Primary and Essential
Includes:	Identify Markings
Extends:	N/A
Cross-refs:	16.1
Use cases:	Identify Markings

Use Case:	Maintain Lane Center
Actors:	Supervisory Control System
Description:	Sends continuous steering commands to steering system to keep vehicles path centered within the detected lane
Type:	Primary and Essential
Includes:	Monitor Lane Position
Extends:	N/A

Cross-refs:	16.2
Use cases:	Monitor Lane Position, Override Control

Use Case:	Overly Complicated Lane Markings
Actors:	Car Sensor Subsystem
Description:	The disabling of LMS based on lane marking visibility
Type:	Primary (essential)
Includes:	N/A
Extends:	The Lane Management System shall be unable to make accurate decisions under the following circumstances, in these circumstances the Lane Management System shall be disabled.
Cross-refs:	3.4
Use cases:	N/A

Use Case:	No Lane Markings
Actors:	Car Sensor Subsystem
Description:	The disabling of LMS based on no lane markings being visible
Type:	Primary (essential)
Includes:	N/A
Extends:	The Lane Management System shall be unable to make accurate decisions under the following circumstances, in these circumstances the Lane Management System shall be disabled.
Cross-refs:	3.5
Use cases:	N/A

Use Case:	Visual Warning Notification on Dashboard
Actors:	Car Sensor Subsystem
Description:	If LMS becomes unable to make decision, notify driver
Type:	Primary (essential)
Includes:	4.2
Extends:	If the Lane Management System is unable to make accurate decisions due to the failure of sensing devices or inaccurate sensing, the Lane Management System shall inform the driver with a visual and auditory alert.
Cross-refs:	4.1
Use cases:	4.2,4.3

Use Case:	Audible Warning Notification on Dashboard
Actors:	Car Sensor Subsystem
Description:	If LMS becomes unable to make decision, notify driver through audible warning
Type:	Primary (essential)

Includes:	4.3
Extends:	If the Lane Management System is unable to make accurate decisions due to the failure of sensing devices or inaccurate sensing, the Lane Management System shall inform the driver with a visual and auditory alert.
Cross-refs:	4.2
Use cases:	4.1,4.3

Use Case:	Visual Diagnostic Notification Warning Notification on Dashboard
Actors:	Car Sensor Subsystem, Driver
Description:	If driver attempts to activate LMS, diagnostic message shall appear
Type:	Primary (essential)
Includes:	N/A
Extends:	If the Lane Management System is unable to make accurate decisions due to the failure of sensing devices or inaccurate sensing, the Lane Management System shall inform the driver with a visual and auditory alert.
Cross-refs:	4.3
Use cases:	4.1,4.2

Use Case:	Visual Warning Notification on Dashboard when LMS is purposely shut down
Actors:	Car Sensor Subsystem, Driver
Description:	If LMS is purposely shut down, LMS shall alert driver with visual notification
Type:	Primary (essential)
Includes:	N/A
Extends:	When the Lane Management System is shut down purposely the Lane Management System shall alert the driver with a visual alert.
Cross-refs:	5.1
Use cases:	N/A

Use Case:	Capture Image
Actors:	Camera Sensor Subsystem
Description:	Captures images to be used by the image processing subsystem in identifying lane markers
Type:	Primary and essential
Includes:	None
Extends:	None

Cross-refs:	Requirement #6
Use cases:	Identify Lane Markers

Use Case:	Record Data
Actors:	Supervisory Control System
Description:	Record decision making data to a collective log including the car's position in the lane, car speed, Lane Management System active status, alerts sent and the times they were sent, and commands sent to the steering system and the times they were sent.
Type:	Secondary
Includes:	None
Extends:	None
Cross-refs:	Requirement #7
Use cases:	None.

Use Case:	Override Control
Actors:	Driver, Supervisory Control Systems
Description:	The driver overrides the Lane Management Systems movement of the steering wheel
Type:	Primary
Includes:	None.
Extends:	Correct lane position
Cross-refs:	Requirement #8
Use cases:	None.

Use Case:	Disable Warning Signal
Actors:	None
Description:	When the turn signal is on, the lane departure warning system should not alert the driver
Type:	Secondary
Includes:	None
Extends:	Alert driver
Cross-refs:	Requirement #9
Use cases:	None

4.2 Domain Model

A domain model was created to define the main elements of the LMS, attributes and major operations of each element, and the structure of connections between the elements. In the model,

system elements are represented by boxes. Attributes and operations pertaining to an element are listed in its corresponding box. Lines between boxes denote some connection between elements, and the text attached to the line describes the nature and multiplicity of the connection. Lines with a black diamond on one end are a type of connection that represent that the element on the flat side of the line is a "part of" the element on the black diamond side. The completed domain model is shown below in Figure 2.

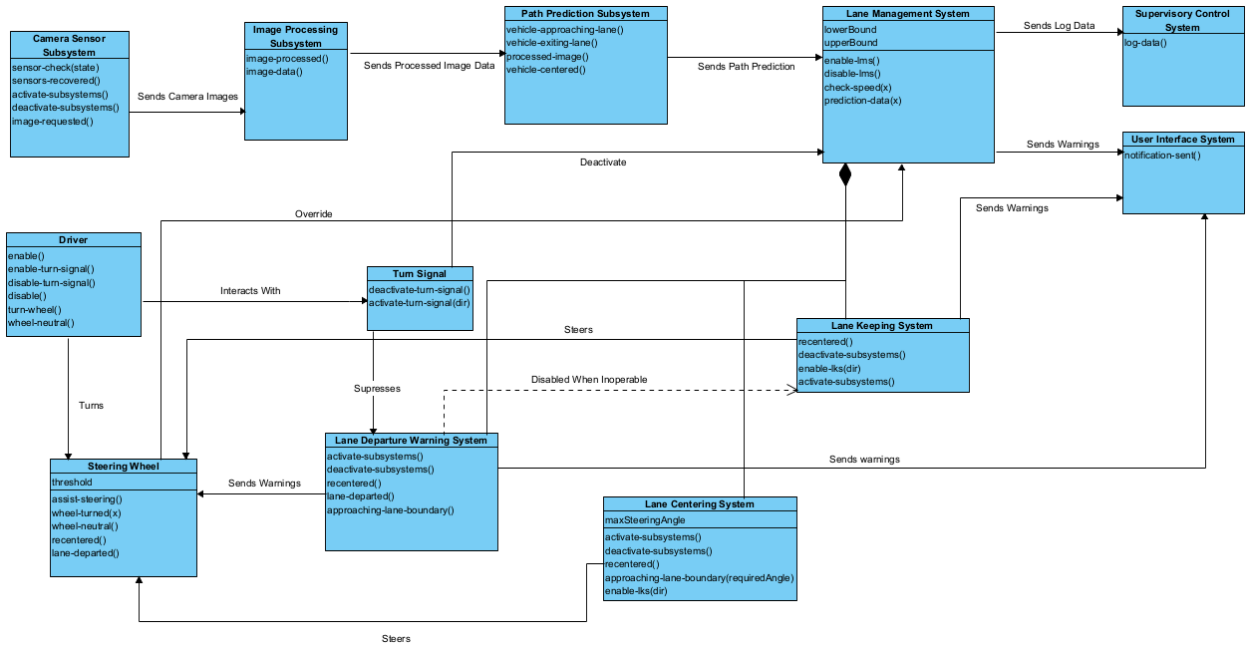


Figure 2: A domain model of the LMS system.

A data dictionary was created to organize the data shown in the domain model. The data dictionary is defined in Table 1 below.

Table 1: Data dictionary defined by domain model from Figure 2.

Element Name		**Description**
Camera Sensor Subsystem		Interface to interact with the Camera Sensors.
Attributes		
Operations		
	sensor-check (state)	Checks if the Camera Sensor Subsystem is in a valid healthy state.
	sensors-recovered()	Restarts the sensor-check if the Camera Sensor Subsystem is believed to have recovered back to a valid healthy state.
	activate-subs	Enables the Camera Sensor Subsystem.

	ystems()	
	deactivate-subsystems()	Disables the Camera Sensor Subsystem.
	image-requested()	Receives an image request and sends images in response to the request.
Relationships		• Sends camera images to the Image Processing Subsystem
UML Extensions		

Element Name		**Description**
Image Processing Subsystem		Processes the images gathered by the cameras and identifies lane markers.
Attributes		
Operations		
	image-data()	Receives images from the Camera Sensor Subsystem to be processed.
	image-processed()	Processes images into data usable by the Path Prediction Subsystem.
Relationships		• Sends processed image data to the path prediction subsystem
UML Extensions		

Element Name		**Description**
Lane Centering System		A component of the Lane Management System. Continuously monitors the vehicle's position within the lane and steers the vehicle to maintain a centered position.
Attributes		
	maxSteering Angle: float	The max angle that the Lane Centering System is able to steering the vehicle at.
Operations		

	activate-subsystems()	Enables the Lane Centering System.
	deactivate-subsystems()	Disables the Lane Centering System.
	recentered()	Returns the Lane Centering System to a monitoring state.
	approaching-lane-boundary(requiredAngle)	Begins the Lane Centering System's steering to recentering the vehicle in the lane.
	enable-lks(dir)	Suspends the Lane Centering System as the Lane Keeping System takes over.
Relationships		• A component of the Lane Management System. • Send steering signals to the steering wheel.
UML Extensions		

Element Name		**Description**
Lane Departure Warning System		A component of the Lane Management System. Warns the driver if they are drifting from the lane they are currently in.
Attributes		
Operations		
	activate-subsystems()	Enables the Lane Departure Warning System.
	deactivate-subsystems()	Disables the Lane Departure Warning System.
	recentered()	Moves the Lane Departure Warning System into an idle state.
	lane-departed()	Sends a message to activate the Lane Keeping System.
	approaching-lane-boundary()	Sends warnings to be displayed.
Relationships		• A component of the Lane Management System. • Sends warning signals to the Steering Wheel.

	• Is able to be suppressed by the Turn Indicator. • Disables the Lane Keeping System when inoperable. • Sends warnings to the User Interface System.
UML Extensions	

Element Name		**Description**
Lane Keeping System		A component of the Lane Management System. Steers the vehicle back into the lane if warning from the Departure Warning System is ignored.
Attributes		
Operations		
	recentered()	Returns the Lane Keeping System to the idle state.
	deactivate-subsystems()	Disables the Lane Keeping System.
	enable-lks(dir)	Starts lane keeping to drive the car back into the middle of the lane.
	activate-subsystems()	Enables the Lane Keeping System.
Relationships		• A component of the Lane Management System. • Sends steering signals to the Steering Wheel. • Is disabled when the Lane Departure Warning System is inoperable.
UML Extensions		

Element Name		**Description**
Lane Management System		The Lane Management System is composed of the Lane Departure Warning System, Lane Keeping System, and Lane Centering System. The Lane management system coordinates interactions between all three subsystems and outside systems.
Attributes		
	lowerBound: Integer	The lowest speed in miles per hour that the Lane Management System will be activated at.

	upperBound: Integer	The
Operations		
	enable-lms()	Enables the LMS system.
	disable-lms()	Disable the LMS system..
	check-speed(x)	Checks if the speed of the vehicle is within the bounds in which the LMS can be active.
	prediction-data(x)	Checks if the prediction data received from the Path Prediction System is valid.
Relationships		● The composite of the Lane Departure Warning System, Lane Keeping System, and Lane Centering System. ● Sends warnings to be displayed to the User Interface System. ● Can be deactivated by the Turn Signal. ● Can be overridden by the Steering Wheel. ● Receives path prediction data from the Path Prediction Subsystem.
UML Extensions		

Element Name		**Description**
Steering Wheel		The steering wheel controlled by the driver
Attributes		
	threshold: float	The amount of torque needed on the steering wheel by the driver to override the Lane Management System.
Operations		
	assist-steering()	Receives steering input from the Lane Management System.
	wheel-turned(x)	Receives steering input from the driver.
	wheel-neutral()	Return the steering wheel to a neutral position.
	recentered()	Stops Steering Wheel haptic feedback.

	lane-departed()	Starts Steering Wheel haptics feedback.
Relationships		• Overrides the Lane Management System. • Receives warning signals from the Lane Departure Warning System. • Receives steering signals from the Lane Keeping System. • Receives steering signals from the Lane Centering System. • Receives steering signals from the driver.
UML Extensions		

Element Name		**Description**
Turn Signal		The left and right turn signal.
Attributes		
Operations		
	activate-turn-signal(dir)	Activates the turn signal.
	deactivate-turn-signal()	Deactivates the turn signal.
Relationships		• Disables the Lane Management System while the turn indicator is active. • Suppresses the Lane Departure Warning System while the turn indicator is active. • Can be activated or deactivated by the Driver.
UML Extensions		

Element Name		**Description**
Path Prediction Subsystem		Predicts the path of the vehicle.
Attributes		
Operations		
	vehicle-appr	Sends messages that the vehicle is approaching a lane

	oaching-lane()	boundary.
	vehicle-exiting-lane()	Sends messages that the vehicle is exiting the lane.
	processed-image()	Processes the image data received from the Image Processing Subsystem in a predicted vehicle path.
	vehicle-centered()	Sends messages that the vehicle is centered in the lane.
Relationships		● Receives processed image data from the Image Processing Subsystem. ● Sends path prediction data to the Lane Management System.
UML Extensions		

Element Name		**Description**
Driver		The driver of the vehicle.
Attributes		
Operations		
	enable()	Enables the Lane Management System.
	enable-turn-signal()	Enables the Turn Signal.
	disable-turn-signal()	Disables the Turn Signal.
	disable()	Disables the Lane Management System.
	turn-wheel()	Sends steering input to the Steering Wheel.
	wheel-neutral()	Stops steering input to the Steering Wheel.
Relationships		● Interacts with the Turn Signal to activate or deactivate it. ● Turns the Steering Wheel.
UML Extensions		

Element Name		**Description**
User Interface System		Interface to give feedback to the user of the vehicle, through audio, haptics, or visuals.
Attributes		
Operations		
	notification-sent()	Displays the sent notification.
Relationships		● Receives warnings from the Lane Management System. ● Receives warnings from the Lane Keeping System. ● Receives warnings from the Lane Departure Warning System.
UML Extensions		

Element Name		**Description**
Supervisory Control System		Saves messages from the Lane Management System into a log.
Attributes		
Operations		
	log-data()	Logs the messages from the Lane Management System.
Relationships		● Receives log data from the Lane Management System.
UML Extensions		

4.3 Sequence Diagrams

Sequence diagrams were made to aid in the visualization of normal and exceptional cases. The following diagrams cover a wide range of system behavior.

Figure 3 shows how the Steering Wheel deactivates the Lane Management System. First the driver turns the Steering Wheel. If the torque of the Steering Wheel is above the threshold to be considered purposeful the Steering Wheel sends a signal to the Lane Management System to deactivate. The Lane Management System distributes this signal to the three Lane Management System subsystems, the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System. All three of these subsystems deactivate in response.

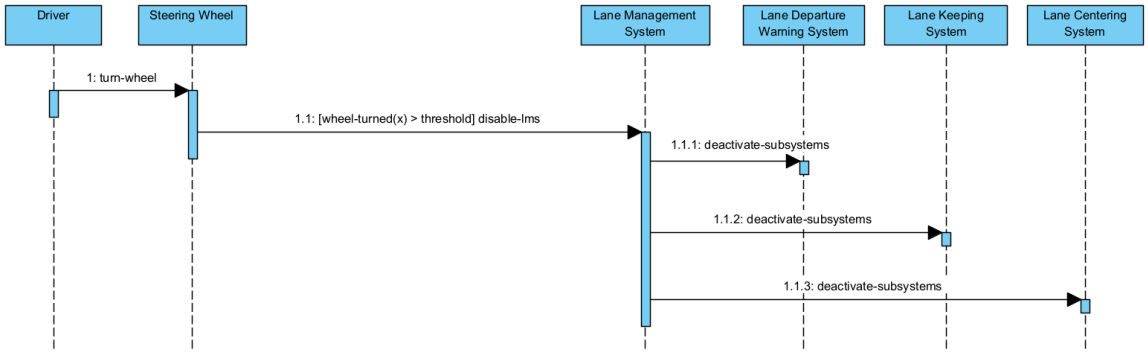


Figure 3: A sequence diagram showing the steering wheel deactivating the Lane Management System.

Figure 4 shows the exceptional case when the steering correction limit has been reached. The sequence starts with the Image Processing Subsystem requesting an image from the Camera Sensor Subsystem. The Camera Sensor Subsystem sends an image in return. This image is sent to the Path Predication Subsystem which determines the road curvature from the image and passes this data to the Lane Centering System. The Lane Centering System will then compute the required steering angle in order for the vehicle to maintain center lane position. The Lane Centering System will attempt to apply steering correction up to the maximum steering angle allowed. If the steering angle limit is reached and the vehicle is still outside of the lane center the Lane Centering System will send a warning to the driver to take control of the vehicle and then deactivate.

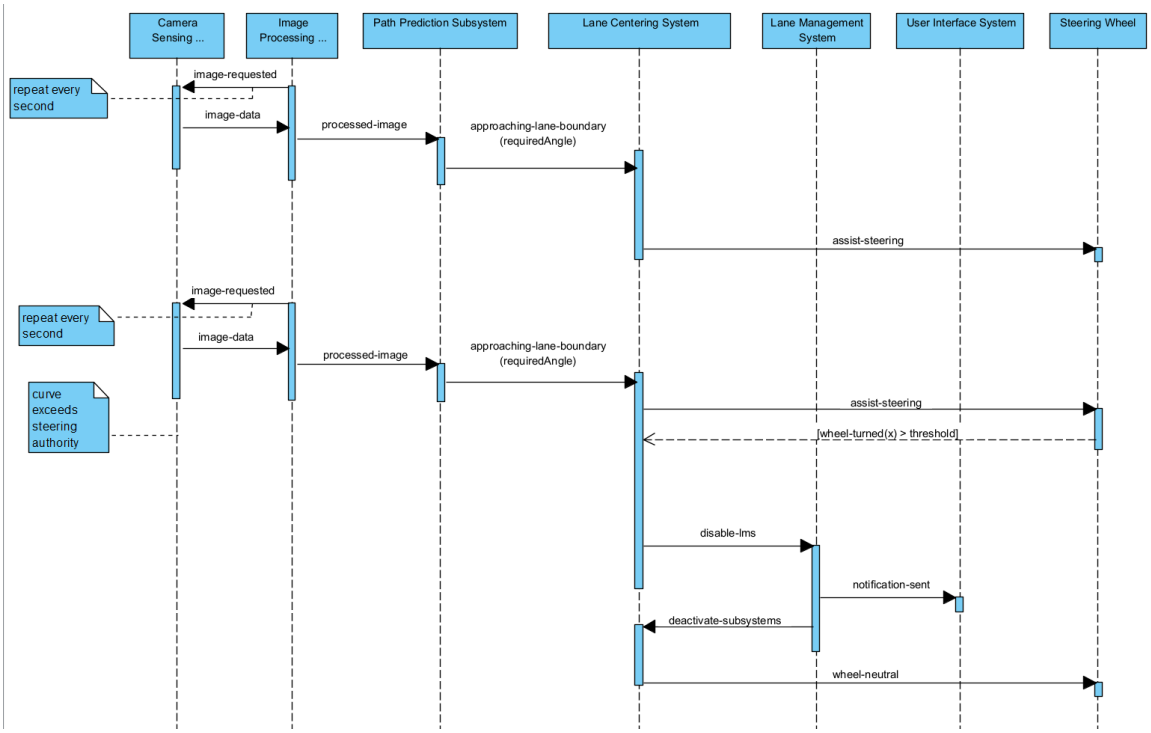


Figure 4: A sequence diagram showing the steering correction limit reached.

Figure 5 shows how the Lane Management System handles seeing multiple lane markings on

the road. The sequence starts by the Image Processing Subsystem requesting an image from the Camera Sensor Subsystem. The Camera Sensor Subsystem returns an image in response. The Image Processing Subsystem determines the lane boundaries from this image and sends them to the Path Prediction Subsystem. The Path Prediction Subsystem attempts to make a vehicle path prediction with the data but due to there being more than expected lane markers the created data is invalid. This predicted path data gets sent to the Lane Management System. The Lane Management System receives the invalid data, due to this expectation the system will deactivate. To deactivate the Lane Management System sends deactivate signals to each of its three subsystems, the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System. All three of these subsystems deactivate in response.

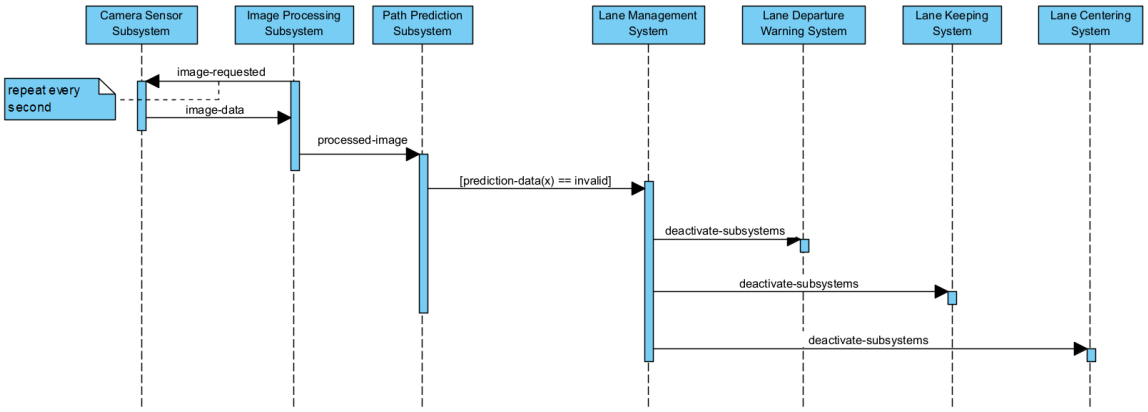


Figure 5: A sequence diagram showing the exception handling of the camera sensors detecting multiple lane markings.

Figure 6 shows how the Lane Centering System handles centering the vehicle in a normal case. The sequence begins with the Image Processing Subsystem requesting an image from the Camera Sensor Subsystem. The Camera Sensor Subsystem sends an image to the Image Processing Subsystem in return. The Image Processing Subsystem processes the image to get the lane marker data and passes this information to the Path Prediction Subsystem. The Path Prediction Subsystem uses the lane marker data to predict the vehicles next position, it then sends this data to the Lane Centering System. The Lane Centering System computes the lane center offset and applies the needed steering correction to keep the vehicle in the center of the lane to the Steering Wheel.

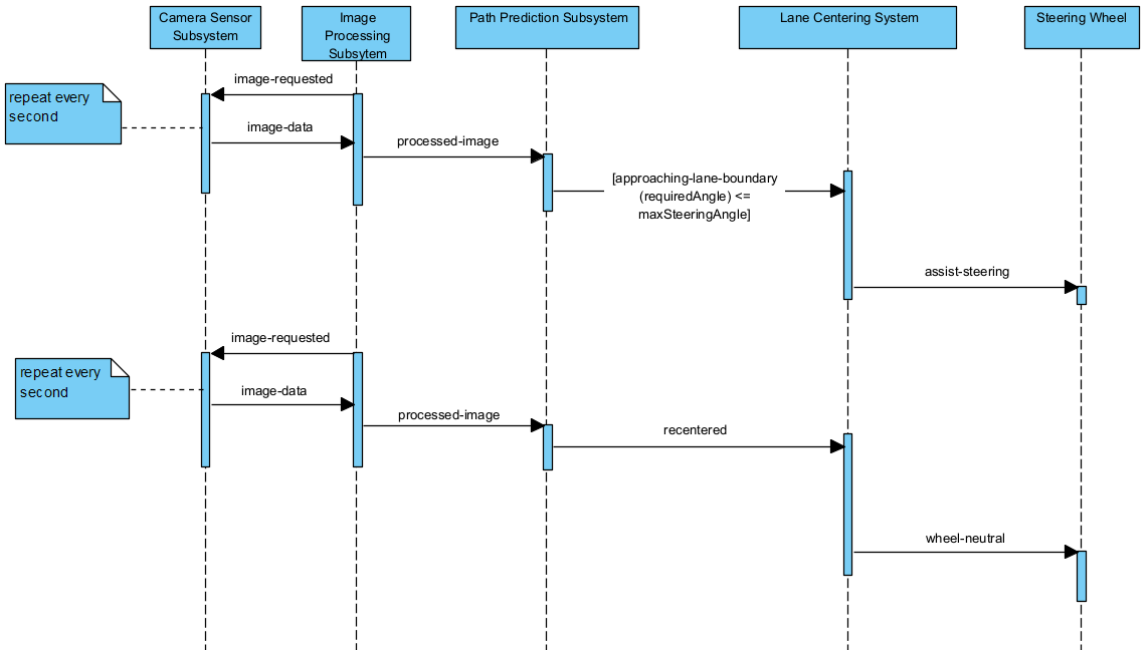


Figure 6: A sequence diagram showing the Lane Centering system maintaining the vehicles center lane position.

Figure 7 shows the Lane Departure Warning System and the Lane Keeping System handling a lane departure. First the Image Processing Subsystem requests an image from the Camera Sensor Subsystem. The Camera Sensor Subsystem sends an image back to the Image Processing Subsystem in return. The Image Processing Subsystem processes the images and sends lane marker data from them to the Path Prediction Subsystem. The Path Prediction Subsystem will either detect a lane departure or not. If a lane departure is detected the Path Prediction Subsystem will signal a departure is detected to the Lane Departure Warning System and the Lane Keeping System. The Lane Departure Warning System will then warn the driver of the departure using the User Interface System. The Lane Keeping System will apply torque to the Steering Wheel as needed to bring the vehicle back into the lane. Once a lane departure is not detected a signal will be sent to the Lane Keeping System to stop torque on the Steering Wheel.

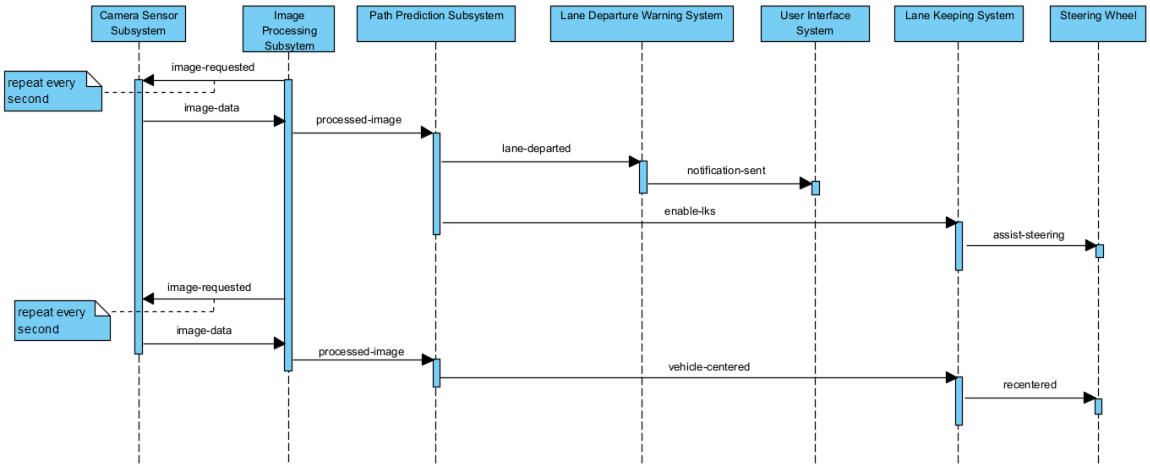


Figure 7: A sequence diagram showing the Lane Departure Warning System and the Lane Keeping Systems handling of a lane departure.

Figure 8 shows how the Lane Management System handles the vehicle being outside of the speed threshold it operates within. The Lane Management System consistently checks if the vehicle speed is within acceptable limits. If the driver's speed is outside of these bounds the Lane Management System will send deactivate signals to all three subsystems, the Lane Keeping System, the Lane Departure Warning System, and the Lane Centering System. In response all three of these systems will deactivate.

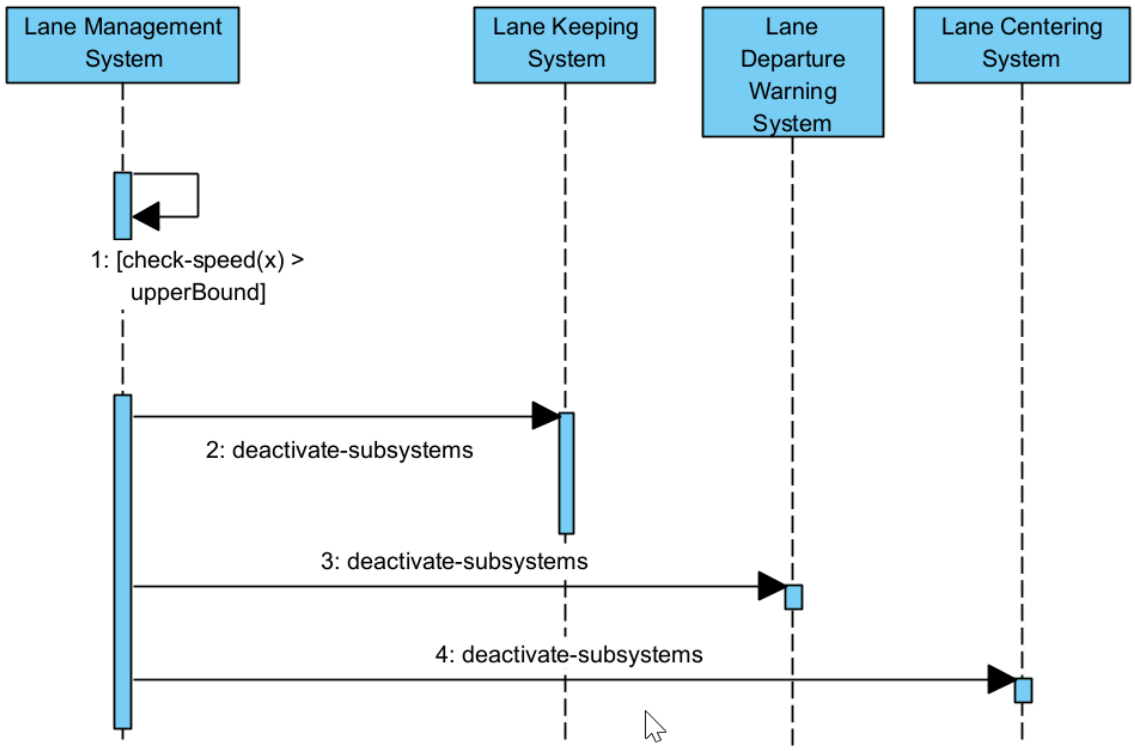


Figure 8: A sequence diagram showing the system handling when the driver is outside of the speed threshold that Lane Management System is active.

Figure 9 shows how the Lane Management System handles no lane markings being detected. The sequence starts by the Image Processing Subsystem requesting an image from the Camera Sensor Subsystem. The Camera Sensor Subsystem then sends camera images to the Image Processing Subsystem. The Image Processing Subsystem sends the image data to the Path Prediction Subsystem. The Path Prediction Subsystem processes the image data and sends the results to theLane Management System. The Lane Management System validates this data. If the data is invalid due to there being no lane markings the Lane Management System sends deactivate signals to all three subsystems, the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System. In response all three of these systems deactivate.

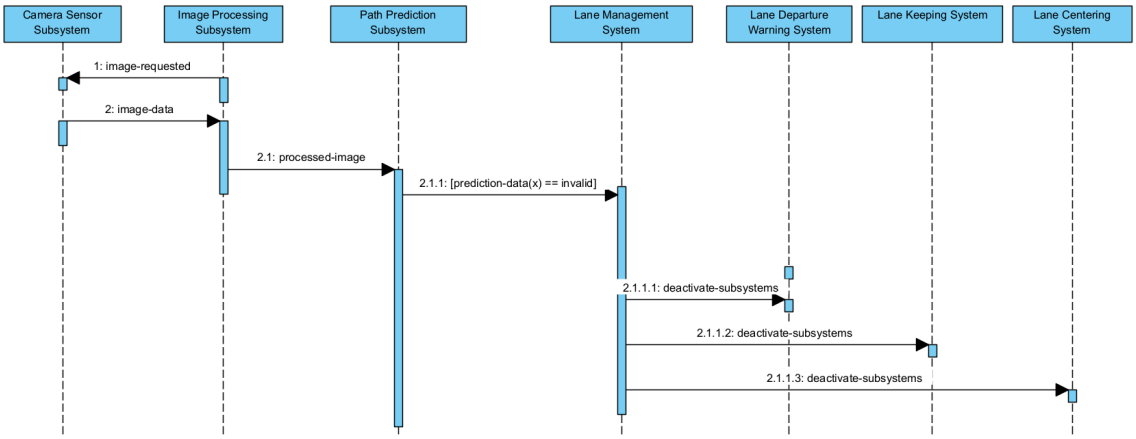


Figure 9: A sequence diagram showing the Lane Management System deactivating due to no lane markings found.

Figure 10 shows the normal case of the Lane Management System capturing and processing images. First the Camera Sensor Subsystem runs camera diagnostics to ensure that the cameras are working properly. Next the Camera Sensor Subsystem captures images of the road. The images are then sent to the Image Processing Subsystem where they are processed to get the lane marking data. This data is sent to the Path Prediction Subsystem. The Path Prediction Subsystem processes the data then sends a message to the Lane Management System containing the processed data.

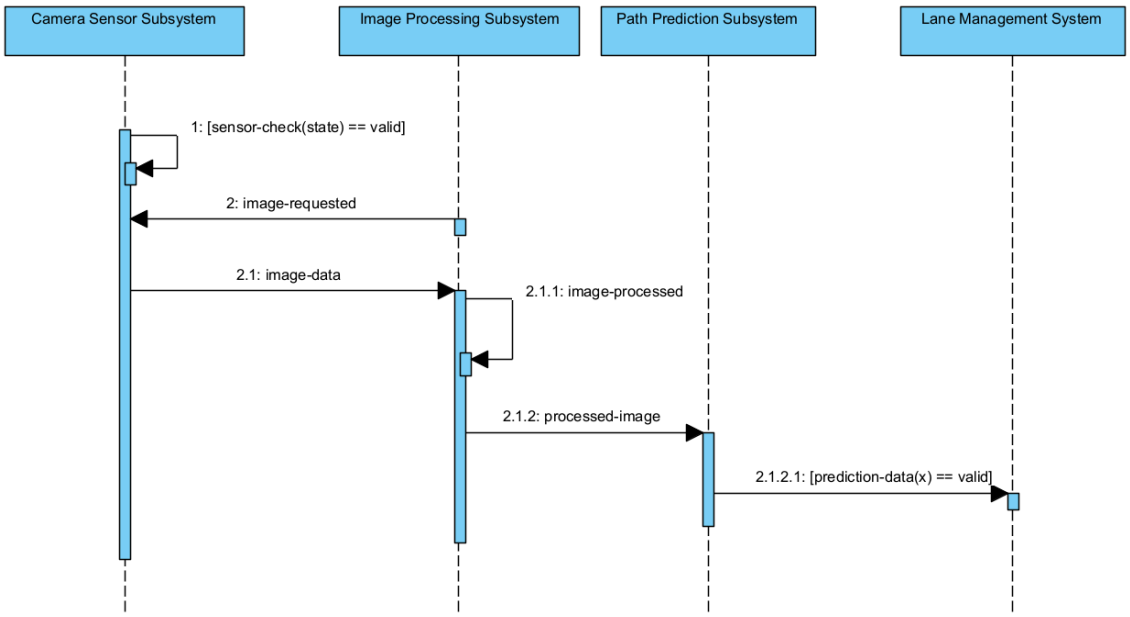


Figure 10: A sequence diagram showing the process of capturing camera data for the Lane Management System to use.

Figure 11 shows the Lane Management System handling the exceptional case where the camera diagnostics fail. First the Camera Sensor Subsystem runs the check sensor diagnostic. The Camera Sensor Subsystem then sends a camera system failure signal back to the Lane Management System. After receiving this signal the Lane Management System deactivates by

sending deactivate signals to its three subsystems, the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System.

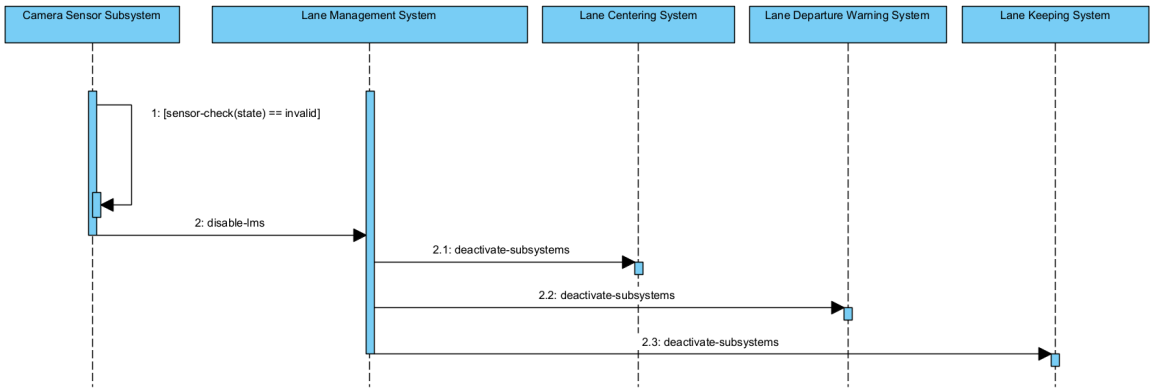


Figure 11: A sequence diagram showing the deactivation of the Lane Management System when the camera sensors fail.

Figure 12 shows how the Lane Management System can be overridden by the Driver using the Turn Signal. First the Driver turns on the Turn Signal. The Turn Signal then sends a message to the Lane Management System that it is active. After receiving this signal the Lane Management System sends deactivate signals to the Lane Keeping System, the Lane Departure Warning System, and the Lane Centering System. These systems deactivate in response to these signals.

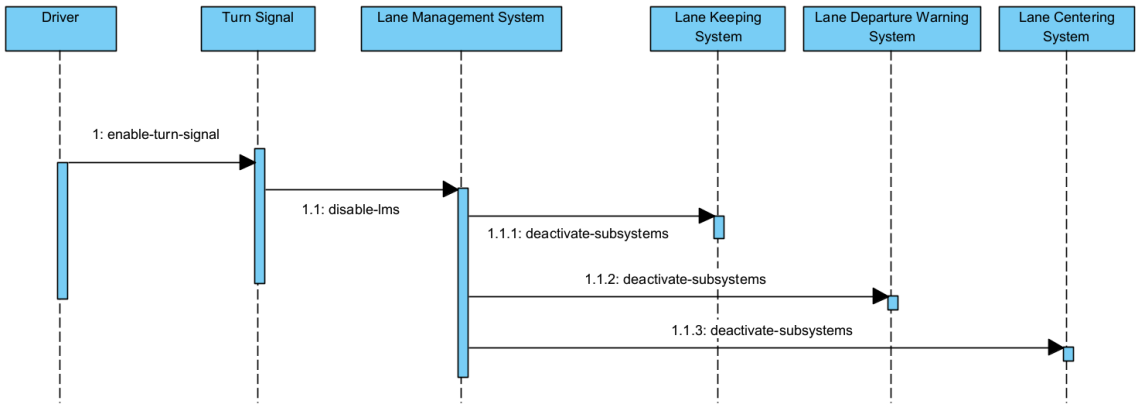


Figure 12: A sequence model showing turn signal deactivating the Lane Management System.

4.4 State Diagrams

Figure 13 shows a state diagram of the Image Processing System. The Image Processing System begins in the request image state where the Image Processing system requests images from the Camera Sensor Subsystem. Once the images are received the Image Processing System moves into the process image state where the images will be processed into usable data. Once the images are processed the Image Processing System will move into the send image data state where the image data will be sent to the Path Prediction Subsystem. Finally the Image

Processing System will wait then transition back to the request image state, restarting the loop of processing.

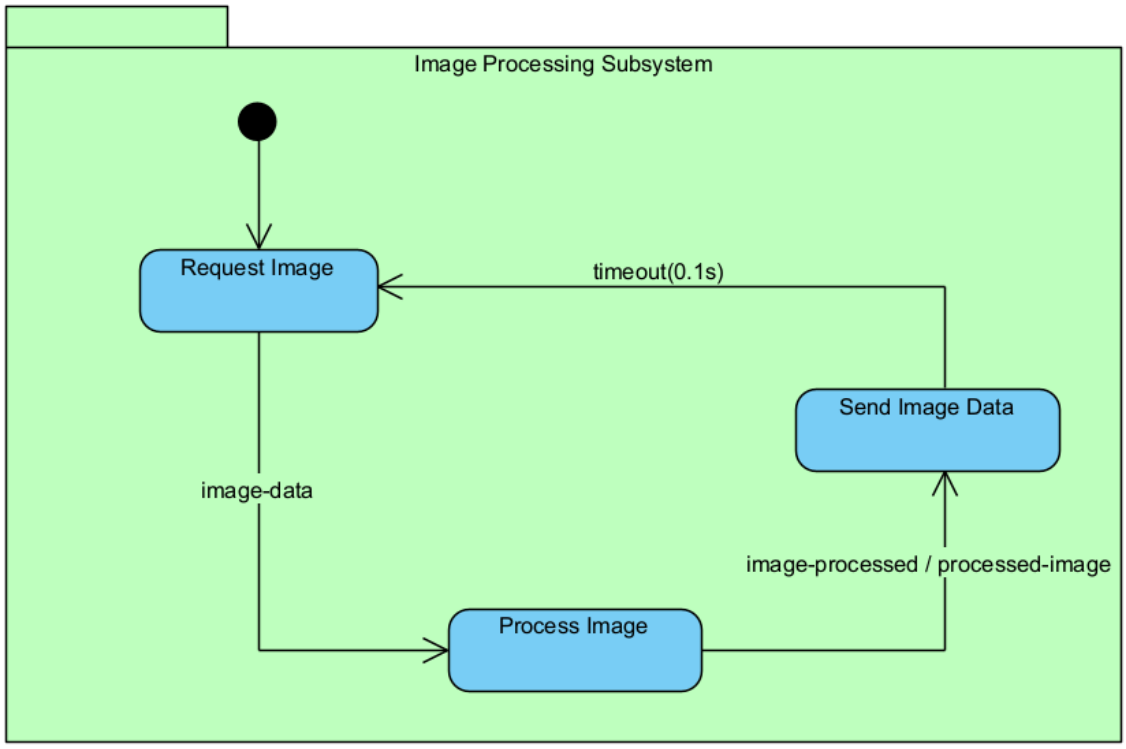


Figure 13: A state diagram of the Image Processing System.

Figure 14 shows a state diagram of the Lane Management System. The Lane Management System begins in the enabled idle state. At intervals of one second the Lane Management System will check the speed bounds to make sure the vehicle is within the upper and lower bounds where the system is allowed to be active. If the vehicle speed is within the bounds the system will transition back to the enabled idle state. If the speed is outside of the bounds the system will transition to the disabled idle state. In the enabled idle state the Lane Management System will receive prediction data from the Path Prediction Subsystem. If this prediction data is valid the system will stay in the enabled idle state, if the data is not valid the system will transition into the disabled idle state. If the Lane Management System is disabled at any time the system will transition into the disabled idle state and disable all related subsystems along with it including the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System. While in the disabled idle state the Lane Management System can be enabled which will additionally send messages to enable all related subsystems including the Lane Departure Warning System, the Lane Keeping System, and the Lane Centering System.

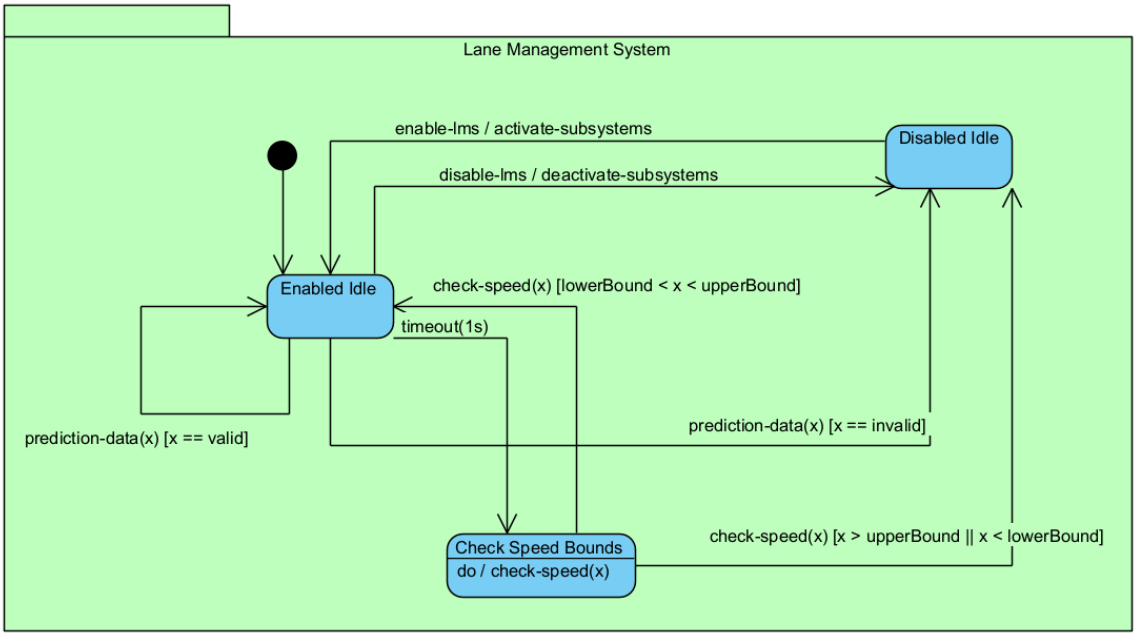


Figure 14: A state diagram of the Lane Management System.

Figure 15 shows a state diagram of the Path Prediction Subsystem. The Path Prediction Subsystem begins in the idle state. Upon receiving processed image data from the Image Processing Subsystem the Path Prediction Subsystem will move into the predicting state. The Path Prediction Subsystem will predict where the vehicle's path is going and classify it as one of three states, approaching lane boundary, lane departure, and centered in lane. The Path Prediction Subsystem will send out a message in respect to which state the predicted path is in. Then the system will return to the idle state.

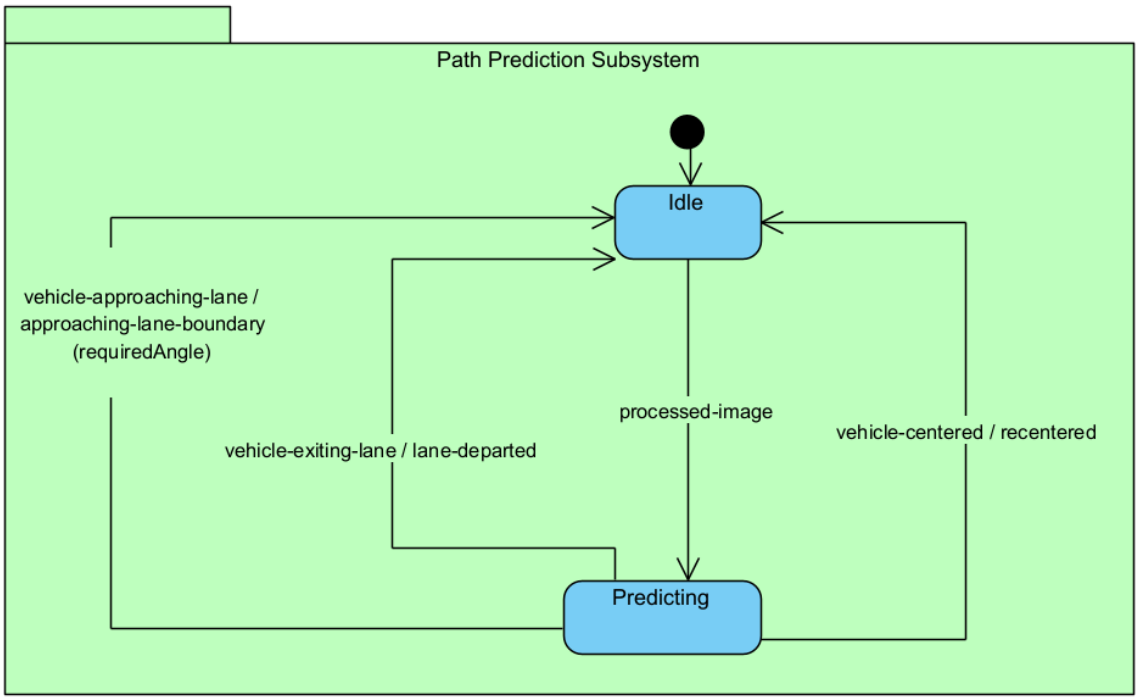


Figure 15: A state diagram of the path prediction subsystem.

Figure 16 shows a state diagram of the LKS. The LKS begins in the idle state. The system will receive a signal to be enabled with the direction lane keeping needs to occur in. Depending if the direction is right or left the system will transition into the keeping: left or keeping: right state. Once the LKS receives a signal that the vehicle is recentered the system will stop lane keeping and return to the idle state. The LKS can be disabled at any time and move into the disabled state. From the disabled state the LKS is able to be enabled.

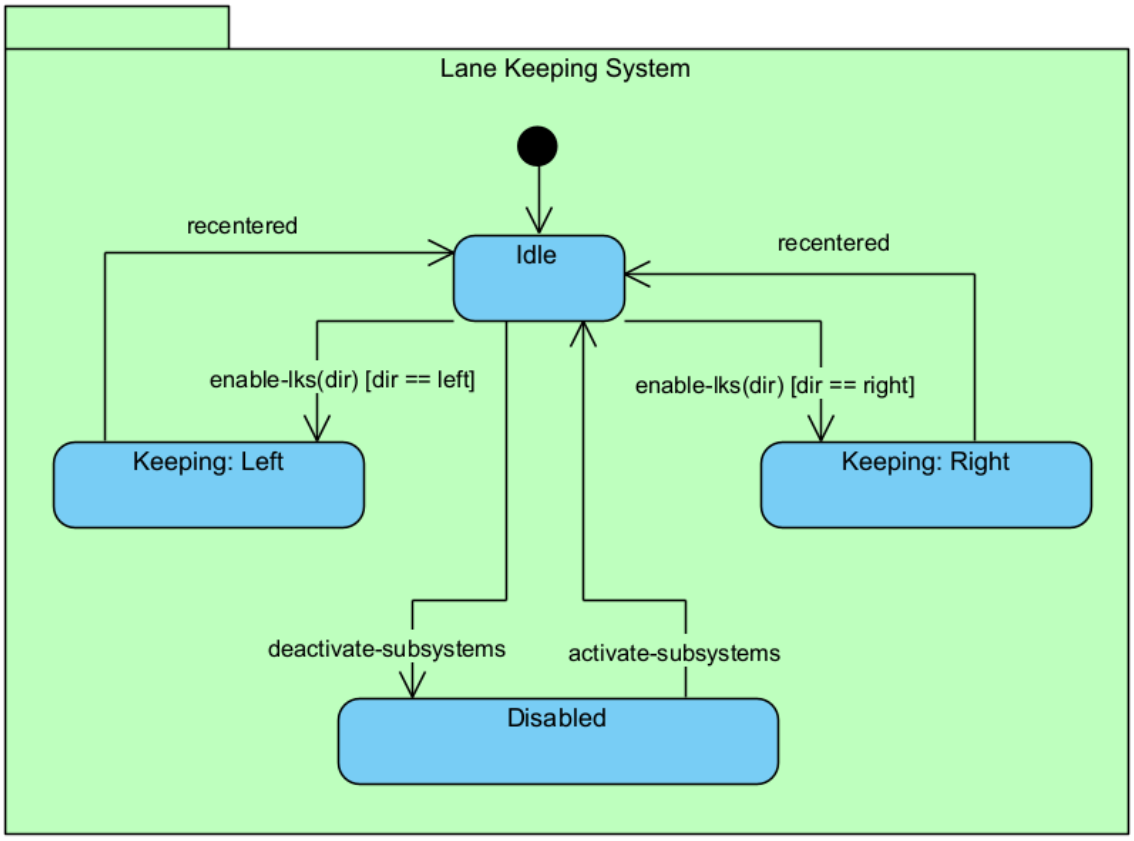


Figure 16: A state diagram of the LKS.

Figure 17 shows a state diagram of the turn signal. The turn signal begins in the Inactive state. The turn signal transitions to the Active: Left state when it is activated with direction = left, and it transitions to the Active: Right state when it is activated with direction = right. From either the Active: Left state or the Active: Right state the turn signal transitions to the Inactive state when it is deactivated.

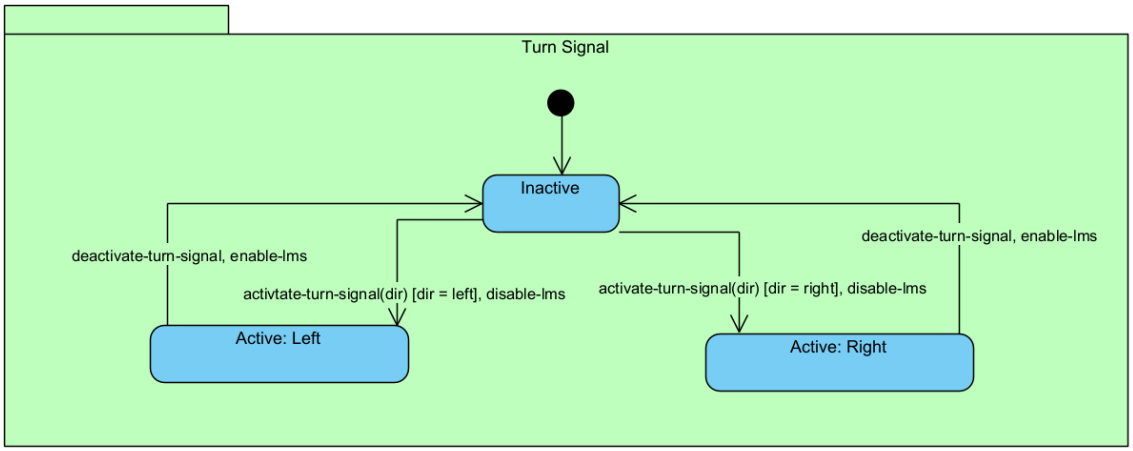


Figure 17: A state diagram of the turn signal.

Figure 18 shows a state diagram of the Lane Departure Warning System. The Lane Departure Warning System begins in the idle state. When the Path Prediction Subsystem indicates that the vehicle is approaching a lane boundary the system moves into the display warning lights state, sending a message to the User Interface Subsystem to display a warning. If the Path Prediction Subsystem indicates that the lane has been departed the system will move into the activate lane keeping system state where an activate message will be sent to the Lane Keeping System. Once the Path Prediction Subsystem indicates that the vehicle has been recentered the Lane Departure Warning System will return to the idle state from either the display warning lights state or the activate lane keeping system state. The Lane Departure Warning System can be disabled at any time moving into the disabled state. When reenabled the Lane Departure Warning System will move back into the idle state.

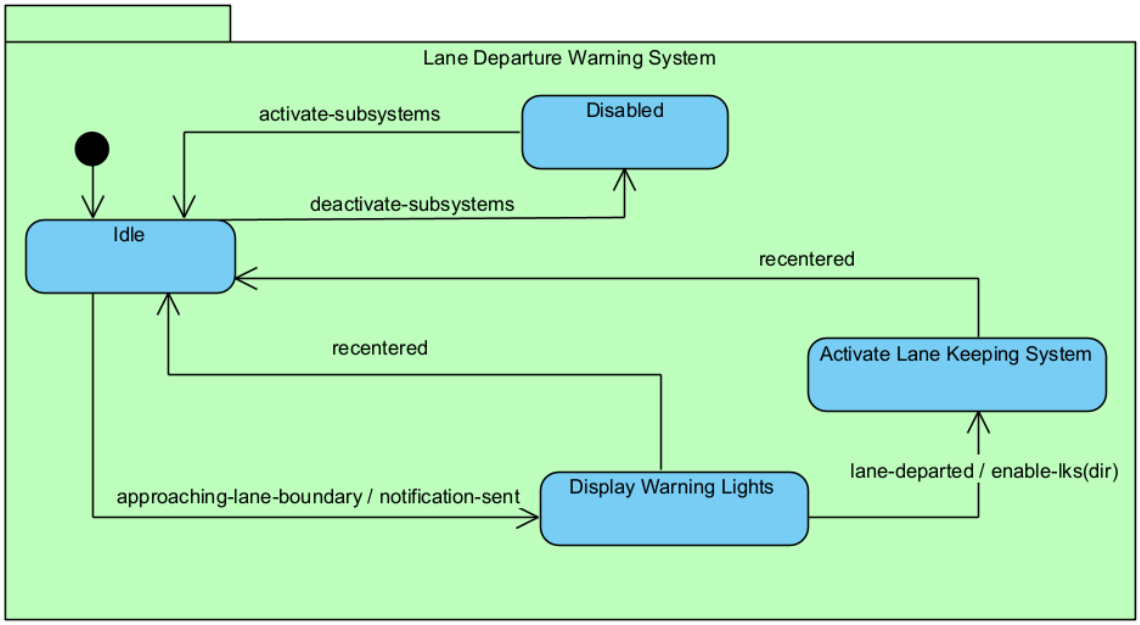


Figure 18: The LDWS state diagram.

Figure 19 shows a state diagram of the User Interface System. It starts by displaying the default interface and stays that way until it receives a notification. The notification feedback is then given for five seconds. After these seconds the User Interface System returns back to its default state.

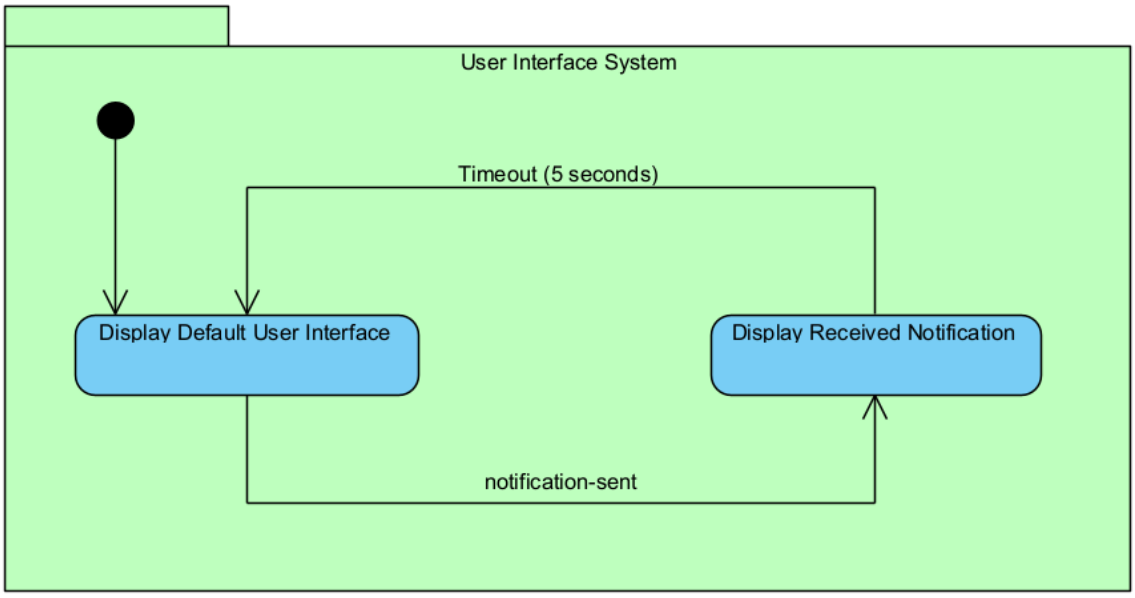


Figure 19: The User Interface System state diagram.

Figure 20 shows a diagram of the Lane Centering System. The Lane Centering System begins in the disabled state. When enabled the system transitions to the monitoring state. The system can be disabled at any time to move back into the disabled state. While in the monitoring state if the Path Prediction Subsystem sends a message that the vehicle is approaching a lane boundary the Lane Centering System will determine if the needed steering angle to recenter is within the maximum angle the Lane Centering System is allowed to steer the vehicle. If the angle needed is lower than the maximum angle the system will move into the centering state and begin sending steering signals to the Steering Wheel. If the needed angle is over the bounds the Lane Centering System will move into the disabled state. Assuming the Lane Centering System entered the centering state, once the vehicle is determined to be recentered by the Path Prediction Subsystem the Lane Centering System will move back into the monitoring state.

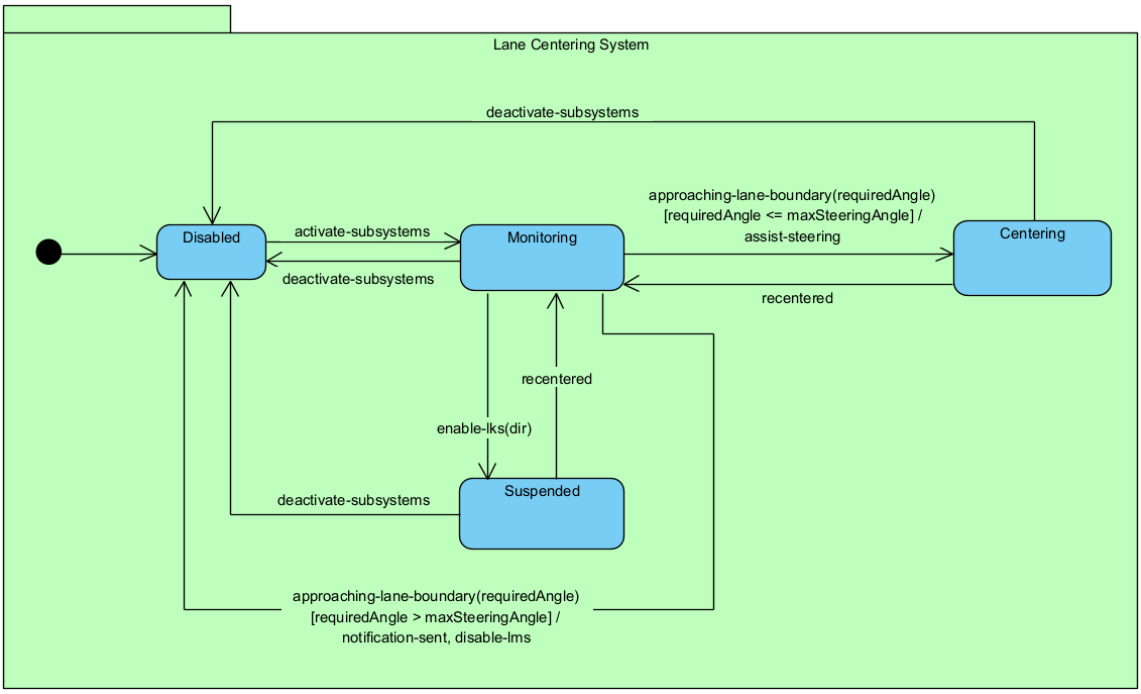


Figure 20: A state diagram of the Lane Centering System.

Figure 21 shows a diagram of the Steering Wheel, which is modeled as two independent state machines because haptic warnings and steering torque do not interact and run in parallel. The steering portion begins in the neutral state. If steering signals are sent from the Lane Keeping System or the Lane Centering System the Steering Wheel will move into the assisted state. Once the vehicle is recentered, meaning the Lane Centering System and the Lane Keeping System no longer need to send steering signals, the Steering Wheel will move back into the neutral state. If at any point the Driver turns the Steering Wheel with more torque than the needed threshold to indicate purposeful steering assisted steering will be overridden and the Driver will take over steering, transitioning the system into the override signaled state. Within the override signaled state if the Steering Wheel returns to a neutral position the system will transition back into the neutral state. Within the haptic feedback state machine the system begins in the not vibrating state. If a lane boundary is departed the system will enter the vibrating state. Once the vehicle is recentered the system will return to the not vibrating state.

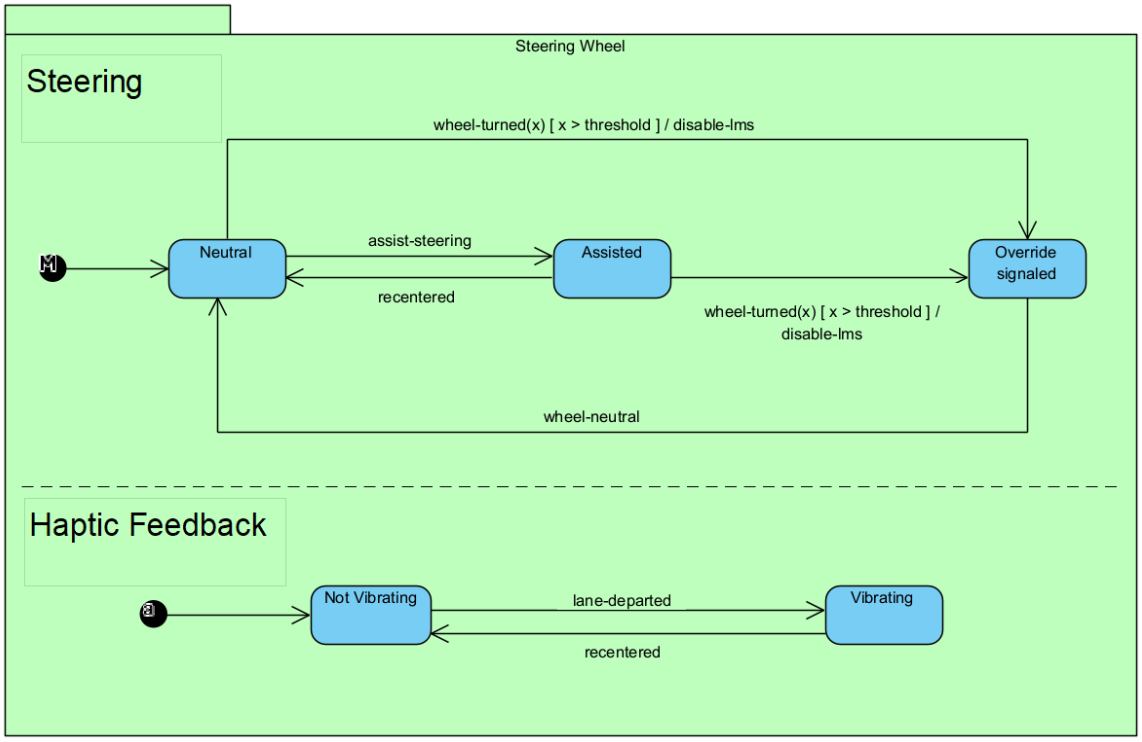


Figure 21: A state diagram of the Steering Wheel.

Figure 22 displays the state diagram of the driver. The Driver begins in the idle state. The Driver can interact with many different systems. The Driver can enable or disable the Lane Management System, turn the Steering Wheel, and enable or disable the Turn Signal. These state transitions will send out messages to the respective systems indicating the action the Driver took.

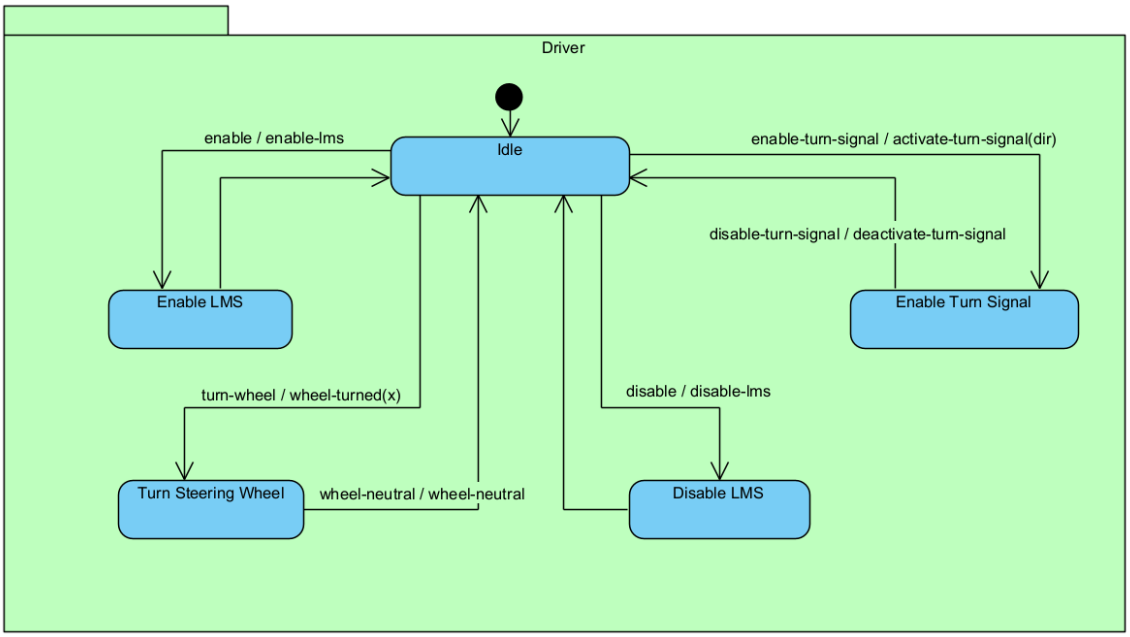


Figure 22: A State Diagram of the Driver.

Figure 23 showcases the state diagram for a Camera Sensor Subsystem. The Camera Sensor Subsystem begins in the sensor initialization check state. If the sensor check returns that the Camera Sensor Subsystem is in a valid state the system will transition to the gather image data state, if not the system will transition into the needs recovery state and send a signal to disable the Lane Management System. From the needs recovery state if the Camera Sensor Subsystem is suspected to be back in a valid state the system will move back into the sensor initialization check state, sending a message to enable the Lane Management System. From the gather image data state when an image is requested from the Camera Sensor Subsystem the image data will be taken and sent to the requesting system then the Camera Sensor Subsystem will return to the gather image data state. The Camera Sensor Subsystem can be disabled at any time moving into the disabled state. If the Camera Sensor Subsystem is reenabled from the disabled state the system will move into the sensor initiation check state.

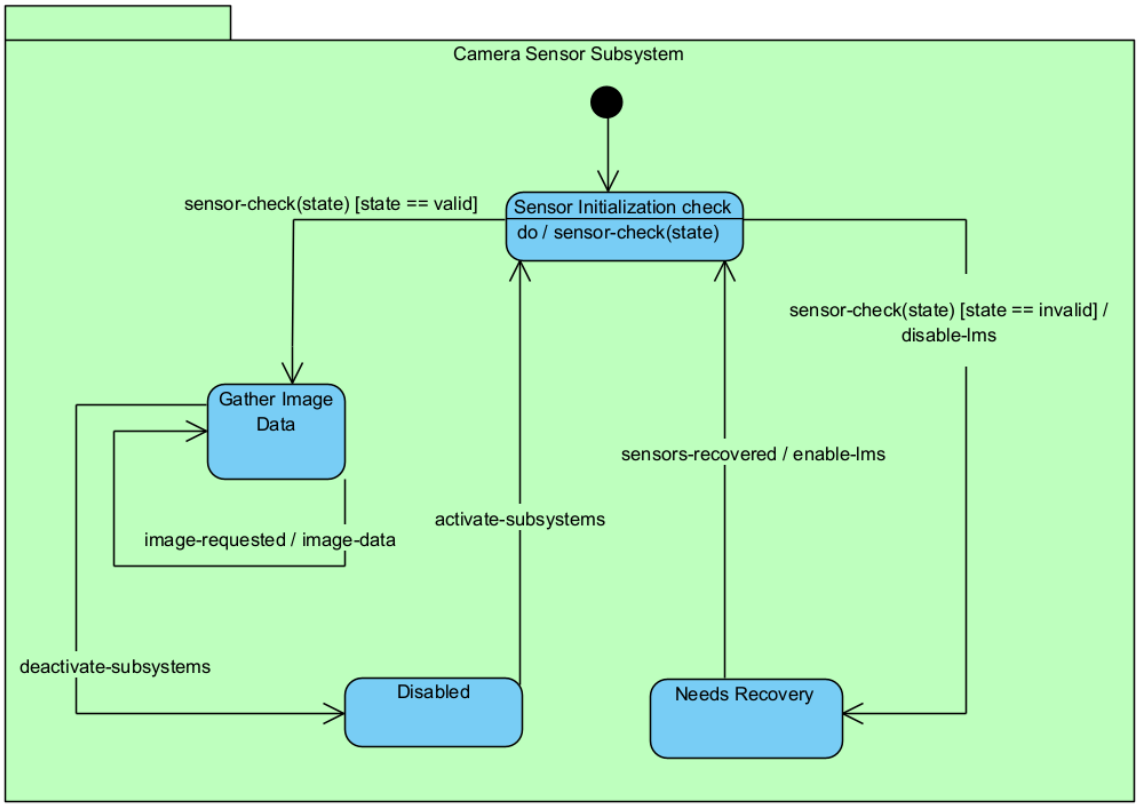


Figure 23: A State Diagram of the Camera Sensor System

5 Prototype

A web-based prototype was created to demonstrate the behavior of the LMS system. The prototype is not the final product, rather it is meant to display how the final product will function.

The prototype has 6 scenarios which follow a scripted sequence of events. Each scenario displays specific conditions that a driver might encounter on the road, and the behavior of the LMS is demonstrated under these conditions. Road and environment variables can be modified to tailor the scenario to different conditions as well.

The prototype also has a free drive mode. In this mode, there is no script and you can toggle any road or environment variable to further test the LMS behavior in conditions not captured by the scripted scenarios.

5.1 How to Run Prototype

The prototype can be run through a browser. To run the prototype, navigate to our team website at https://www.egr.msu.edu/cse435/2026/lms1/. Navigate to the "Links" section of the homepage, and find the "Project" subsection. There should be a link titled "prototype" which will take you to our team's prototype.

Once in the prototype there are instructions in the display window to interact with its features. You can modify environment and road variables yourself, start any of the 4 scripted scenarios, or enter free drive mode.

5.2 Sample Scenarios

There are 6 scenarios in the prototype which follow a scripted sequence of events to demonstrate the behavior of the LMS under conditions a driver might experience on the road. This section details each of those scenarios.

5.2.1 Scenario 1: Ideal Road

Figure 24 displays scenario 1, LMS behavior on an ideal road in first person point-of-view. Throughout this scenario the road is completely straight and lane markers are fully visible. The scenario starts with the driver accelerating into the speed threshold and activating LMS. After, the driver veers out of their lane which prompts the LKS to steer the vehicle back within lane boundaries.



Figure 24: Image of prototype scenario 1 from first person POV.

5.2.2 Scenario 2: Curves

Figure 25 displays scenario 2, LMS behavior on a road with curves from a chase point-of-view. As the scenario progresses curves are generated on the road with differing radii and in different directions. Throughout the scenario, lane markers are fully visible. The initial speed in the scenario starts at 60 mph but drops to 40 mph in order for the LMS to easier handle tighter curves.

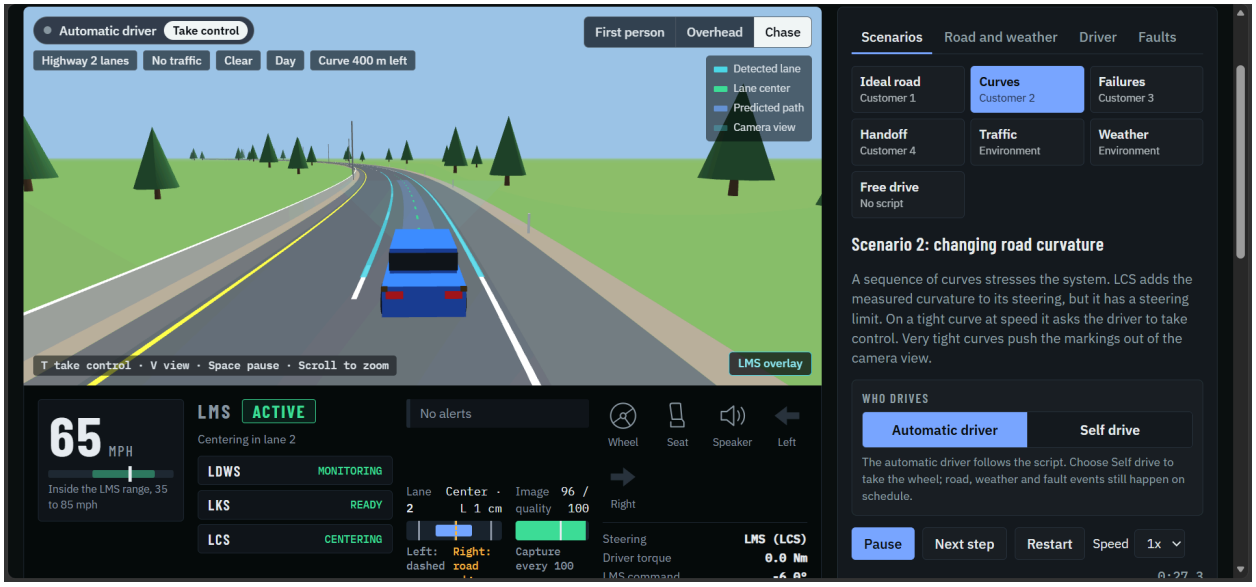


Figure 25: Image of prototype scenario 2 from chase POV.

5.2.3 Scenario 3: Failures

Figure 26 displays scenario 3, LMS behavior while handling edge cases or experiencing system failures. In the image, the LMS deactivates because too many lane markers are detected.

By default, the road is straight throughout the entire scenario. Errors handled during this scenario include: obscured camera, too many lane boundaries detected, lane boundaries faded or obscured, lane too narrow, failure of multiple LMS subsystems, and erroneous, forged commands.



Figure 26: Image of prototype scenario 3 from chase POV.

5.2.4 Scenario 4: Control Handoff

Figure 27 displays scenario 4, LMS behavior during control handoff between LMS and the driver. Throughout the scenario, the actor steering the vehicle changes between the LMS and driver due to turn signal activation, driver steering with enough torque to override LMS, and driver manually deactivating LMS.

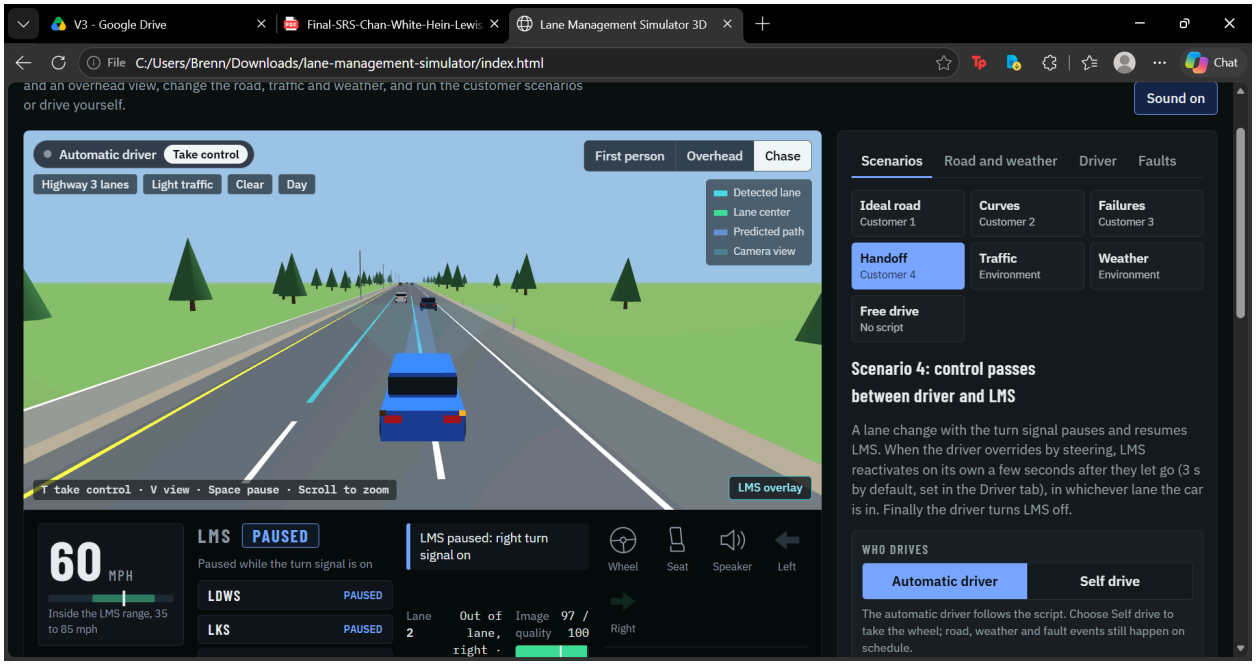


Figure 27: Image of prototype scenario 4 from chase POV.

5.2.5 Scenario 5: Traffic

Figure 28 displays scenario 5, LMS behavior with traffic on road. The presence of other vehicles on the road has no effect on LMS behavior, however it's useful to show what LMS can't do. The LMS retains all of its functionality from previous scenarios, however control is taken by the driver to avoid other vehicles on the road. By default, the road is completely straight throughout the scenario.



Figure 28: Image of prototype scenario 5 from chase POV.

5.2.6 Scenario 6: Weather

Figure 29 displays scenario 6, LMS behavior under varying weather conditions. Throughout the scenario the weather conditions change between the following: clear, raining, foggy, and snowing. By default, the road is completely straight throughout the scenario.

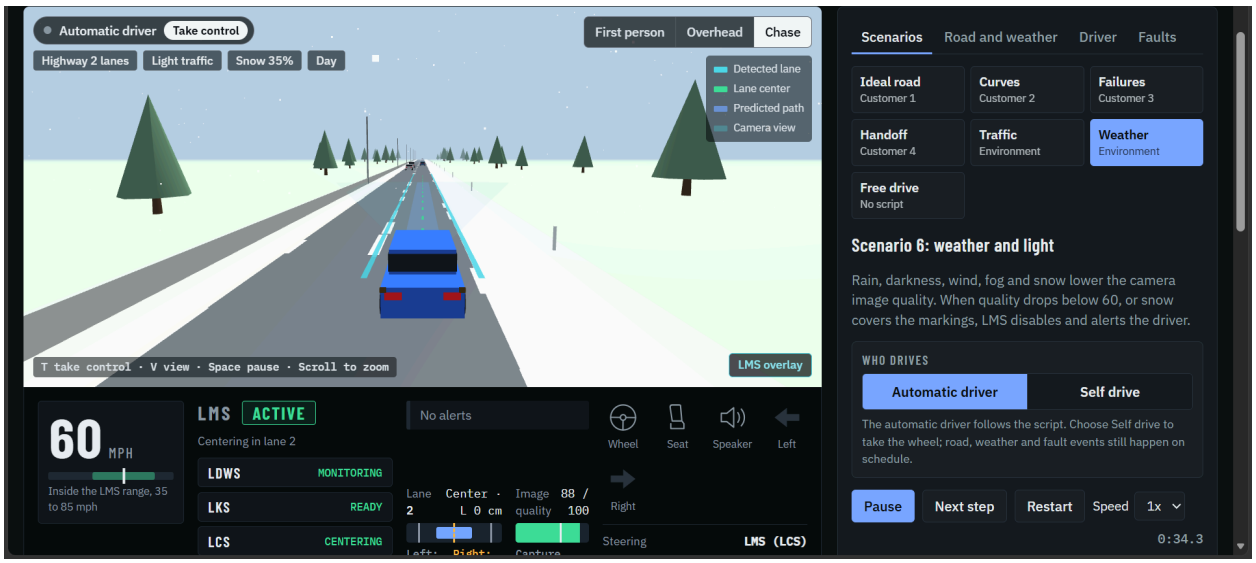


Figure 29: Image of prototype scenario 6 from chase POV.

6 References

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7 Point of Contact

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